

Marco Acea

About

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Programming Languages

Python

C++

TypeScript

JavaScript

Software/Frameworks

OpenAI API

🤗 Hugging Face

PyTorch

scikit-learn

pandas

Dask

LLM Prompt Engineering

Distributed Systems

Bloomberg API

Alpaca

Algorithmic Trading

Risk Management

Real-time Systems

Feature Engineering

Production ML

Education

Carnegie Mellon University

08/2018 - 12/2022

Bachelor's of Science in Electrical and Computer Engineering

- Structure and Design of Digital Systems
- Logic Design and Verification
- Web App Development
- Intro to Embedded Systems
- Principles of Imperative Computing
- Fundamentals of Embedded Control

Experience

Bloomberg L.P.

07/2024 - Present

Software Engineer – Electronic Trading Order Management Systems

- Contributed to a global electronic trading order management system used by thousands of financial institutions, supporting millions of trades and order events daily
- Led development of an A/B testing and analysis framework that evaluated 10,000+ samples and identified high-impact issues before affecting live trading workflows
- Built distributed logging and tracing tools to pinpoint system latency bottlenecks, reducing critical execution delays by up to 5+ seconds and improving trader responsiveness
- Developed intuitive analytics dashboards that allowed trading desks, product teams, and engineers to audit system behavior and performance within seconds rather than hours

Boeing

03/2023 - 04/2024

Software Engineer

- Developed automated UI integration tests using Selenium for web applications which significantly improved code base test coverage
- Developed Virtual Machine devices for the High Fidelity Digital Twin Program

Intuit

05/2021 - 08/2022

Software Engineering Intern - DevX, Argo Project

- Developed a containerized GraphQL Proxy Service using Typescript, Docker, and Redis for optimization, that protects API services from potentially malicious incoming requests using rate limiting and static query analysis
- Developed new feature for the ArgoCD project that allows non-admin users to safely create Applications and Projects for themselves within a Kubernetes cluster.

23andMe

07/2020 - 09/2020

Software Engineering Intern

- Built Flask-based internal application management dashboard for the Application Infrastructure Team, streamlining service monitoring and authentication workflows.

Projects

A/B Testing Risk Analysis Platform

10/2025 - Present

Unsupervised Learning A/B Testing Quantitative Analysis Equities

- Built a Python-based A/B testing analysis platform used to analyze large-scale behavioral data and extract actionable signals from 10,000+ experimental samples
- Used unsupervised pattern discovery to identify non-obvious anomalies and structural deviations across experiments before client impact
- Identified 5 critical integration issues and 3 runtime anomalies, reducing rollout risk by 20% across experiments affecting thousands of users globally

Mid-Frequency ML Trading System

Ongoing

Algorithmic Trading Risk Management Real-time Systems Feature Engineering

- Built electronic trading system processing S&P 500 data with neural networks (294K+ parameters) achieving 60%+ model accuracy and sub-100ms signal generation for real-time execution.
- Conceived and developed production-grade system with risk management, position sizing, and automated model retraining, featuring real-time P&L tracking and order execution via Alpaca API.
- Implemented monitoring stack with Prometheus/Grafana tracking win rates, fill rates, model/system latency metrics across 500+ symbols with 50+ technical